

TF

TRAINING

Science of Hypertrophy

12-Week Muscle Building Program

50 Pages

■10,000 Value

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Chapter 1: Three Mechanisms of Muscle Growth

Muscle hypertrophy occurs through three interconnected mechanisms. All three are stimulated by resistance training and should be periodically targeted for maximal growth.

Mechanism	Definition	Best Stimulated By	Training Variable to Maximise
Mechanical Tension	Force generated by active muscle contraction under load	Heavy compound lifts, full ROM, controlled tempo	Load $\geq 65\%$ 1RM, progressive overload
Metabolic Stress	'The pump' — lactate/H ⁺ /Pi accumulation in muscle	High rep sets (12–20), short rest, blood flow restriction	Volume, short rest 60–90s, BFR training
Muscle Damage	Micro-tears in myofibrils triggering repair & growth	Eccentric emphasis (lowering phase), new exercises	3–4s eccentric tempo, novel exercises

Volume Landmarks — Sets Per Muscle Group Per Week

Muscle Group	MEV (Min Effective)	MAV (Max Adaptive)	MRV (Max Recoverable)	Beginner Start
Chest	10 sets	14–18 sets	22 sets	8–10 sets
Back (width)	10 sets	14–18 sets	22 sets	8–10 sets
Back (thickness)	10 sets	14–18 sets	22 sets	8–10 sets
Shoulders	8 sets	12–16 sets	20 sets	6–8 sets
Biceps	8 sets	12–16 sets	20 sets	6–8 sets
Triceps	8 sets	12–16 sets	20 sets	6–8 sets
Quads	10 sets	16–20 sets	26 sets	8–10 sets
Hamstrings	8 sets	12–16 sets	20 sets	6–8 sets
Glutes	8 sets	14–18 sets	22 sets	6–8 sets
Calves	8 sets	14–18 sets	24 sets	6–8 sets

Muscle Group	MEV (Min Effective)	MAV (Max Adaptive)	MRV (Max Recoverable)	Beginner Start
Abs/Core	8 sets	12–16 sets	20 sets	6–8 sets

12-Week Hypertrophy Program — Push Pull Legs

Day	Session	Exercise	Sets × Reps	Rest	RPE Target
Mon (Push)	Chest/Shoulders/Triiceps	Barbell Bench Press	4×6–8	3 min	8–9
		Incline DB Press	3×10–12	2 min	8
		Cable Flyes	3×13–15	90s	8
		Seated DB OHP	4×8–10	2 min	8
		Lateral Raise	4×15–20	60s	9
		Skullcrusher	3×10–12	90s	8
Tue (Pull)	Back/Biceps	Deadlift	4×4–6	4 min	8–9
		Weighted Pull-ups	4×6–10	3 min	8
		Seated Cable Row	3×10–12	2 min	8
		Lat Pulldown	3×12–15	90s	8
		Barbell Curl	3×10–12	90s	8
		Hammer Curl	3×12–15	60s	8
Wed (Legs)	Quads/Hams/Glutes	Barbell Back Squat	4×6–8	4 min	8–9
		Leg Press	3×10–12	2 min	8
		Bulgarian Split Squat	3×10–12/side	2 min	8
		Romanian Deadlift	4×8–10	2 min	8
		Leg Curl	3×12–15	90s	8

Day	Session	Exercise	Sets × Reps	Rest	RPE Target
		Standing Calf Raise	4×15–20	60s	9

Progressive Overload Methods

Method	How to Apply	When to Use	Example
Load Progression	Add 2.5–5kg when top rep range achieved	Primary method for beginners	Hit 3×10 at 60kg → try 62.5kg
Rep Progression	Add 1–2 reps per set each week	When stuck on weight increases	3×8 → 3×9 → 3×10 → increase weight
Volume Progression	Add 1 set per muscle group every 2 weeks	Intermediate lifters	3 sets → 4 sets over 4 weeks
Density Progression	Same weight/reps in less time	For conditioning + hypertrophy	3×10 in 30 min → same in 25 min
Tempo Manipulation	Slow eccentric 3–4 seconds	Plateau breaking	3 seconds down on every rep
Technique Refinement	Improve ROM and muscle activation	Always — never stop improving	Full depth squat vs partial